

**Name:** I Forgot Again Team **Date:** 4/23/24 **Class:** Engineering

# MyDesign Portfolio

## Element I Initial Template

Data Table:

|   | requirement                                                                 | testing setup and procedure                                                                                                                                                                                                                                                                                                                                                                                                              | test cases and number of trials | expected range of results                                                                                                                                                                  | pass/fail criteria, goal                                                                                                                               |
|---|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | The prototype must have a proper fit into the car console.                  | Materials: Ruler, car, car console, and prototype. <ol style="list-style-type: none"> <li>1. Measure the length, width, and height of the prototype with the ruler.</li> <li>2. Measure the length, width, and height of the car console with the ruler.</li> <li>3. Place the prototype into the car console.</li> <li>4. Record if the measurements of the device fit in the car console and if the lid iteration can open.</li> </ol> | 1                               | Even with an iteration, the prototype does not fit.<br>The prototype does not fit but it can fit with an iteration.<br>The prototype successfully fits into the car console and is stable. | Pass:<br>The prototype successfully fits into the car console and is stable.<br>Fail:<br>The prototype fails to fit into the car console.              |
| 2 | The prototype must keep the items intact and undamaged while in the device. | Materials: Car keys, phone, prototype, and car (NOTE: Not all cars can be stored in the device as they are required to be in the key holder of the vehicle). <ol style="list-style-type: none"> <li>1. Place both the phone and car keys into the device and close it.</li> <li>2. Go on a 30-minute drive (can be any route).</li> <li>3. Once stopped, record if</li> </ol>                                                            | 1                               | The prototype and the items placed are both dented or damaged in some way.<br>The prototype is dented or damaged in some way but the items inside are intact.                              | Pass:<br>Both the prototype and the items in the prototype are undamaged and intact.<br>Fail:<br>The prototype and the items placed are both dented or |

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|   |                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                  | <p>the condition of the items.</p> <ol style="list-style-type: none"> <li>Damage to items should be calculated using a physics gravitational equation to see how far the item has been dropped in the prototype. <math>\text{Force} = \text{Gravity} \times \text{distance between object 1} \times \text{distance between object 2} / \text{Distance between those objects squared}.</math></li> <li>Repeat this test any amount of times in different cars for results.</li> </ol>                                                               |   | <p>The prototype is intact but the items inside are dented or damaged in some way. Both the prototype and the items in the prototype are undamaged and intact.</p>                                                                             | <p>damaged in some way. The prototype is dented or damaged in some way but the items inside are intact. The prototype is intact but the items inside are dented or damaged in some way.</p>                                                |
| 3 | <p>The prototype must successfully hold items so that the driver can focus their full attention on the road.</p> | <p>Materials: Car, car keys, phone, and prototype. (NOTE: Not all cars can be stored in the device as they are required to be in the key holder of the vehicle).</p> <ol style="list-style-type: none"> <li>Place both the phone and car keys into the device and close it.</li> <li>Go on a 30-minute drive (can be any route).</li> <li>After the drive, record if the driver is distracted while driving. The driver should self-reflect and notice any changes to their emotions.</li> <li>Use the Driver Safety Measurement System</li> </ol> | 1 | <p>The driver is constantly worrying about their items and is driving distracted. The driver sometimes worries about the items and is driving distracted. The driver only worries about the items a normal amount and drives undistracted.</p> | <p>Pass: The driver only worries about the items a normal amount and drives undistracted. The driver barely has to worry about their items and drives undistracted. (The driver must be safe, even if an accident wasn't their fault).</p> |

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|  |  |                                                                                                                                                |  |                                                                                                                                             |                                                                                                                                                                 |
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|  |  | (DSMS) Methodology to determine if the driver was distracted or not.<br>5. Repeat this test any amount of times in different cars for results. |  | The driver barely has to worry about their items and drives undistracted (The driver must be safe, even if an accident wasn't their fault). | Fail:<br>The driver is constantly worrying about their items and is driving distracted. The driver sometimes worries about the items and is driving distracted. |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Test 1: Proper Fit

The design should accommodate various car models and measurement configurations to ensure the safety of people and objects. This test is a high priority and involves using geometric measurements and a ruler to measure the fit of the prototype into a car console. Note that the dimensions of the prototype are 10 inches x 5 inches x 6.5 inches. To set up the test, you need a ruler, car, car console, and the prototype. First, measure the prototype's length, width, and height with the ruler. Second, measure the car console's length, width, and height with the ruler. Third, place the prototype into the car console. Finally, record if the device's measurements fit in the car console and if the lid iteration can open. This test should be done once but can be performed on any number of cars or be repeated. The range of results for one car includes the prototype doesn't fit, the prototype doesn't fit but can with an iteration, or the prototype successfully fits into the car console and is stable. In order to pass, the prototype must successfully fit into the car console and stay stable. To fail, the prototype will not fit into the car or doesn't currently but could be with an iteration. Our goal is to have the prototype remain stable and fit into the console.

### Results:

The prototype was 10x5x6.5 inches and fit into a 10.5x6x10 inch space. This means that our prototype passed the test because it was able to fit within the constrained space.

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## Test 2: Durability

The items placed in the prototype must be fully intact during, and after the vehicle is in motion. This test is a high priority and uses physics through a gravitational equation ( $\text{Force} = \text{Gravity constant} \times \text{Mass of object 1} \times \text{mass of object 2} / \text{distance}$ ) to calculate if damage has been done to the items in the prototype. Car keys, a phone, the prototype, and a car are needed for this test. Not all cars can be stored in the device as they are required to be in the key holder of the vehicle, so this can be replaced with any other, similar-sized item. First, place both the phone and car keys into the device and close it. Second, go on a 30-minute drive on any route. Third, once stopped, record the condition of the items. Finally, damage to items should be calculated using a physics gravitational equation to see how far the item has been dropped in the prototype ( $\text{Force} = \text{Gravity constant} \times \text{Mass of object 1} \times \text{mass of object 2} / \text{distance}$ ). This test should be performed once but can be done any number of times with different cars. Also, the test measures if the item has been dented or damaged. The range of results includes the prototype and the items placed are both dented or damaged in some way, the prototype is dented or damaged in some way but the items inside are intact, and the prototype is intact but the items inside are dented or damaged in some way, or both the prototype and the items in the prototype are undamaged and intact. For the prototype to pass, both the prototype and the items in the prototype are undamaged and intact. Otherwise, the prototype fails if the prototype and the items placed are both dented or damaged in some way, the prototype is dented or damaged in some way but the items inside are intact, or the prototype is intact but, the items inside are dented or damaged in some way. Our goal is to have both of our items in the prototype to be safe.

Results: There was no damage done to the items in the prototype, therefore, no force was applied and the force equation was equal to 0.

## Test 3: Safety

The driver must be focused on driving and not the location of their objects. In other words, their items must be secure so that the driver can focus. This test is a high priority for the prototype and uses engineering design and safety principles with the Driver Safety Measurement System (DSMS) methodology which can show if drivers are being safe

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on the road. The materials for the test include a car, car keys, a phone, and the prototype (not all cars can be stored in the device as they are required to be in the key holder of the vehicle). First, place both the phone and car keys into the device and close it. Second, go on a 30-minute drive (can be any route). Third, after the drive, record if the driver is distracted while driving. The driver should self-reflect and notice any changes to their emotions. Finally, use the Driver Safety Measurement System (DSMS) Methodology to determine if the driver was distracted or not. This test can be performed once but can be done any number of times with different cars. The driver could be constantly worrying about their items and is driving distracted, sometimes distracted, distracted at a normal amount, or undistracted. The driver must be safe, even if an accident wasn't their fault. For the prototype to fail, The driver is constantly worrying about their items and is driving distracted or the driver sometimes worries about the items and is driving distracted. To pass, the driver only worries about the items a normal amount and drives undistracted or the driver barely has to worry about their items and drives undistracted. Our goal is to have our prototype result lower the amount of distractions while driving to keep others safe.

Results: According to the Driver Safety Measurement System, our drive placed us in the safest category of drivers, meaning the prototype passed.

### Data Conclusion:

To conclude, our prototype was successful and is practical because it passed all of the tests for measurements, durability, and safety.

|                   |  |  |  |  |  |  |
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| <b>Element I:</b> |  |  |  |  |  |  |
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| Testing,<br>Data<br>Collection<br>and Analysis                                                                                                                                                               |                          |                           |                              |                                   |                          |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------|------------------------------|-----------------------------------|--------------------------|---------------------|
|                                                                                                                                                                                                              | 5                        | 4                         | 3                            | 2                                 | 1                        | 0                   |
| <b>1. Quantity of tests conducted.</b> The student conducts _____ test<br>s for high priority requirements that are reasonable based on instructional context, or through physical or mathematical modeling. | <i>more than four</i>    | <i>four</i>               | <i>three</i>                 | <i>two</i>                        | <i>one</i>               | <i>no</i>           |
| <b>2. Importance of tests.</b> The student conducts tests for _____ requirements that are reasonable based on instructional context, or through physical or mathematical modeling.                           | <i>all high priority</i> | <i>most high priority</i> | <i>half of high priority</i> | <i>some low and high priority</i> | <i>only non priority</i> | <i>no</i>           |
| <b>3. Comprehension of testing procedure.</b> The student demonstrates _____                                                                                                                                 | <i>considerable</i>      | <i>ample</i>              | <i>adequate</i>              | <i>partial or overly general</i>  | <i>minimal</i>           | <i>a failure of</i> |



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|                                                                                                                                                                               |                                                                                                     |                                                                                                 |                                                                                                        |                                                                                                                               |                                                                                          |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------|
| understanding of testing procedure, including the gathering and analysis of resultant data.                                                                                   |                                                                                                     |                                                                                                 |                                                                                                        |                                                                                                                               |                                                                                          |           |
| <b>4. Detail of the design analysis.</b><br>The analysis of the effectiveness with which the design met stated goals includes<br><hr/> explanation (and summary) of the data. | <i>a consistently detailed (generously supported by pictures, graphs, charts and other visuals)</i> | <i>a generally detailed (generally supported by pictures, graphs, charts and other visuals)</i> | <i>a somewhat detailed (at least somewhat supported by pictures, graphs, charts and other visuals)</i> | <i>a partial (partially complete and/or partially correct and minimally supported by pictures, graphs, and other visuals)</i> | <i>an attempted (may not be supported by pictures, graphs, charts and other visuals)</i> | <i>no</i> |